

A NOTE ON MECHANISTIC MEAN ESTIMATION FOR A TRAINED WIDE MLP WITH POLYNOMIAL ACTIVATIONS

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ABSTRACT. We give a mechanistic algorithm that competes with Monte Carlo sampling for mean estimation in wide multilayer perceptrons after a finite number of online SGD training steps. The network is initialized with Gaussian weights and trained on fresh Gaussian inputs. Our starting point is the mean and cumulant-propagation framework of Wu et al. [3] for untrained random networks. We show how the structure exposed and written by training can be recorded causally and propagated through the estimator, yielding a train-and-explain analogue of their sampling-competitive approach. For fixed training horizon, we obtain quantitative error and runtime guarantees. We also obtain a nonquantitative growing-horizon result for a training horizon ($T(n) \rightarrow \infty$), although explicit quantitative control of such growing horizons remains open. This is a preliminary note: we give the full algorithm together with proof sketches and heuristic explanations of the main claims, rather than complete proofs.

1. INTRODUCTION

This note is motivated by the questions raised by the ARC team in their blogs [1, 2]. The agenda asks whether a structural or mechanistic information about a neural network can be used to estimate its output distribution more efficiently than a black-box estimation. In [3], Wu et al. provides a cumulant propagation algorithm with estimates on floating point-operations runtime. The motivation for this project was the question whether mean and covariance propagation¹ alone can compete with Monte Carlo sampling for trained networks. The main claim of this note is that this is possible under certain constraints that we will explain later in this note; see Theorem 1.1.

In [2], the ARC team explains about an “advice” π that should train/update along with the neural network, which can provide information to the mean estimator. There are conditions on what this advice can be, which is laid out in [1]. The condition of being “mechanistic” is still loosely defined, nevertheless, we believe that the advice we develop (see (3.4)) in this note is ARC style mechanistic. The algorithm (see pseudocode 6) stemmed out of trying to understand Wu et al.[3] and trying to formalize the philosophies and ideas in [1, 2]; for example see Sections 3.1, 4. So far, we haven’t tested out our algorithm (or a variant) on other types on neural networks. But, it seems likely that any progress on a randomly initialized untrained network can be pushed to a trained network as well (at least for fixed training time).

Now, we state the main theorem.

Theorem 1.1 (Informal main theorem). *Fix the depth L , polynomial activations $\phi_1, \dots, \phi_L$, a learning rate $\eta > 0$ independent of n , and a training horizon $T \geq 1$. Consider a width- n Gaussian-initialized MLP trained for T steps of online SGD on fresh Gaussian inputs against a Gaussian-regular target (see Section 2 for details). There exists a causal online advice process $(\pi_t)_{t \leq T}$ (see (3.4)) such that the final advice π_T has size²*

$$O_L(nT + T^2)$$

that guides an estimator $\mathbb{G}_M(W_T, \pi_T)$, which estimates the mean $\mu_T := \mathbb{E}_{x \sim \mathcal{N}(0, \mathbb{I}_n)}(M_{W_T}(x))$ satisfying

$$\sup_{1 \leq i \leq n} \mathbb{E} [|(\mathbb{G}_M(W_T, \pi_T))_i - (\mu_T)_i|^2] \leq \frac{\text{polylog}_{L,T}(n)}{n}.$$

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¹Our algorithm should also be able to handle higher cumulants but we don’t expound it in this note.

²The cost of maintaining the entire advice up to time T is $O_L(nT^2 + T^3)$.

If D^3 is as in Section 5.1, the tensorized thin feature dimension is $O_L((1+T)^D)$, and the estimator can be implemented in

$$O_L(n^2(1+T)^D + n(1+T)^{2D} + (1+T)^{3D})$$

arithmetic operations. In particular, the leading dense-matrix cost is

$$O_L(n^2(1+T)^D),$$

and for fixed L and T the total runtime is $O_{L,T}(n^2)$.

Remark 1.2. For a fixed time T , one can in fact get the runtime to be $O_L(n^2T + nT^2 + T^3)$ for the same error bound in Theorem 1.1 using perturbative methods and completely avoiding tensorization technique from Section 5.1. So, this raises the question “why tensorization?”. Indeed, in this regime the training-created off-diagonal correlations are perturbatively small, so the higher Hermite terms are already lower order. Its advantage is instead that it removes the first-Hermite truncation on the explanation-selected covariance: for polynomial activations, the entire matched-Gaussian nonlinear image of the thin covariance is propagated exactly. Thus the additional $(1+T)^D$ computational cost buys robustness to larger training-created correlations, which becomes relevant when one tries to let the training horizon grow with n .

Remark 1.3. The present estimates are not uniform enough in T to give an explicit growing training horizon. Nevertheless, the theorem for every finite T implies by a diagonal argument that there exists some deterministic $T(n) \rightarrow \infty$ for which the error is $n^{-1+o(1)}$ and the running time is $n^{2+o(1)}$.

2. THE SETTING

2.1. Initialization. The untrained setting is same as in [3]. We consider a fully connected MLP with width $n \in \mathbb{N}$ and depth $L \in \mathbb{N}$ ⁴, and polynomial activation functions $\phi^{(\ell)}$ for a layer (ℓ) . The weight matrix for a layer $1 \leq l \leq L$ is denoted by $W_0^{(\ell)} \in \mathbb{R}^{n \times n}$. By abuse of notation we sometimes drop the superscript (ℓ) and denote the weight matrix for a generic layer as W_0 . We will now describe the initialization for these weights. Let

$$W_0 = G, \quad G_{ij} \sim \mathcal{N}(0, 1/n) \text{ iid},$$

i.e., the weights are Gaussian initialized.

2.2. Training. We will perform an ordinary forward/backward online SGD. We choose a target family of functions $f_n : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with the following control such that $\exists K, \beta$ independent of n , such that for every coordinate $i \in \{1, \dots, n\}$ and $u \geq 0$ and $x \sim \mathcal{N}(0, \mathbb{I}_n)$

$$\mathbb{P}(|f_n(x)|_i > u) \lesssim \exp[-(u/K)^\beta].$$

We call this a Gaussian regular condition. Essentially, the idea is to impose a good enough control such that we get uniform bounded moments of every fixed order and, by a union bound, $\|f_n(X)\|_\infty$ being $O(\text{polylog}(n))$ with arbitrarily high polynomial probability. During training (we use a normalized square loss function), say after time t , each layer is accessed by operations of the form

$$h \mapsto W_{t-1}h \text{ (forward step),} \quad d \mapsto W_{t-1}d^\perp \text{ (backward step)}$$

where $h, d \in \mathbb{R}^n$, and W_{t-1} is the weight matrix after $t-1$ training steps for $1 \leq t \leq T$. In this note, we don’t get into the nitty-gritty of precise definitions of $h^{(\ell)}$ and $d^{(\ell)}$ for each layer l using online SGD.

Since, we are performing an online SGD, each W will get a rank one write from the training given by

$$W \leftarrow \frac{\eta}{n} dh^\perp,$$

where $\eta > 0$ is the learning rate that is independent of n ⁵.

³Think of D as the product of degrees of the polynomial activation functions.

⁴Assume $n \gg L$ as we wish to let $n \rightarrow \infty$ while fixing a finite L . For an untrained network we follow Wu et al. [3] and hence we suppress any asymptotic estimates in L .

⁵If we wish to train indefinitely as n grows, then one will probably have to impose some restrictions of the form $\eta(t) \sim 1/t$

2.3. The goal. Now, let M_{W_T} denote the above discussed wide MLP after T training steps as discussed above. The goal is to estimate

$$\mathbb{E}_{x \sim \mathcal{N}(0, \mathbb{I}_n)}(M_{W_T}(x)),$$

efficiently in comparison with Monte Carlo sampling.

3. MECHANISTIC ADVICE FROM TRAINING TRANSCRIPT

In the beginning of this section we will first decompose the weight matrices $W_T^{(\ell)}$ into different amenable parts. Then, using these decompositions we will define a “mechanistic advice”. Again, we will drop the superscript (ℓ) to denote a generic layer.

3.1. Structure vs Randomness. In [3][Section 7.2], Wu et al. mentions that one may need to adopt a “structure vs random”⁶ approach to do a mechanistic estimation of the mean. Inspired by this idea, we do a structure vs randomness split that we will describe now.

Let B_T denote all the right-query (forward pass) exposed up to time T , and A_T is the span of all left-query (backward pass) directions. Let $P_{A^\perp}$ and $P_{B^\perp}$ denote the corresponding orthogonal projections. Observe that the rank of P_A, P_B is $O(T)$. So, if $T \ll n$, P_A, P_B are very low rank matrices compared to W_T . Hence, we do the following split.

$$(3.1) \quad W_T = \underbrace{S_T}_{\text{structure}} + \underbrace{P_{A^\perp} G P_{B^\perp}}_{\text{randomness}},$$

where G is an $n \times n$ Gaussian matrix. Here S_T is measurable with respect to the chronological training transcript, while the matrix G appearing in the orthogonal complement can be taken conditionally fresh Gaussian.

Remark 3.1. In fact S_T also has rank $O(T)$ and hence can be seen as a perturbation if n is large. So, one can show that covariance propagation of Wu et al. [3] can be done (i.e. $O(n^3)$ run time for $O(1/n^2)$ MSE) even for a trained network using perturbative analysis if one is not concerned about the asymptotic bounds in T . Moreover, it seems unlikely that such perturbative analysis will be less effective if one is concerned about extending the techniques developed in this note to a wider class of NNs.

One can make the “diffusiveness” of S_T, P_A, P_B rigorous using the following observation

$$(3.2) \quad \|S_T\|_{\max}, \|P_A\|_{\max}, \|P_B\|_{\max} \lesssim O(\text{polylog}_T(n)/n),$$

where $\|\cdot\|_{\max}$ is the entry wise max norm and polylog_T denotes that the constants and the powers in the polylog can depend on T .

Remark 3.2. The importance and technicality of proving (3.2) is understated here. It is the first non-trivial and the most important engine of the entire algorithm we are describing in this note.

Remark 3.3. The bottleneck of letting $T(n) \rightarrow \infty$ as $n \rightarrow \infty$ is (3.2). This is one of the reasons one will have to get a decay for the learning rate as T increases.

Now, we introduce another important idea that will improve⁷ our algorithm. In the blog [2][Footnote 9.], the ARC team mentions that maintaining and propagating all covariances using Wu et al. [3] algorithm might be costly and hence “the advice” could tell us what important covariances to keep track of. This motivates another decomposition of the structured part S_T that we will describe now. To see where exactly this decomposition helps us, see Section 4.

Observe that

$$\Delta W := -\frac{\eta}{n} dh^\top,$$

⁶This ergodic theory style idea is one of my main motivation to study mechanistic mean estimation for trained networks.

⁷The improvement doesn’t exist for a fixed T in the sense of FLOP, but definitely in the clock time. Moreover, if eventually, we want to let $T(n) \rightarrow \infty$ as $n \rightarrow \infty$, the new decomposition should definitely help.

can be absorbed into S_T after a training step since h, d have already been exposed to training⁸. Consequently, after T training steps, S_T is a sum of only $O(T)$ rank-one matrices. In particular, we can get a factorization

$$(3.3) \quad S_T = P_T R_T^\top,$$

where P_T, R_T have $O(T)$ number of columns.

3.2. The advice. In order to maintain the decompositions (3.1), (3.3) after every training step, we need to keep track of the relevant information that will allow us to do that. Hence we need an “advice”.

First, for each later l , we keep track of an orthonormal bases $Q_{A,T}^{(\ell)}$ and $Q_{B,T}^{(\ell)}$ for the exposed left and right query spaces respectively. Moreover, we also keep track of $P_T^{(\ell)}, R_T^{(\ell)}$ for (3.3). Hence, for a layer (ℓ) , we take the advice schematically to be

$$(3.4) \quad \pi_T^{(\ell)} = \{Q_{A,T}^{(\ell)}, Q_{B,T}^{(\ell)}, P_T^{(\ell)}, R_T^{(\ell)}\}.$$

It is not hard to see that for a fixed depth L , the size of the training advice $|\pi_T|$ is

$$O_L(nT + T^2).$$

Moreover, this is the sense in which we believe that π_T is mechanistic in the ARC style: it records the directions in which the training has causally broken the original Gaussian symmetry; it does not ask which training directions look statistically important, rather, the training itself selects them.

4. COMPRESSING THE COVARIANCE

In this section, we will describe the covariances we will propagate through the linear part of a layer using the advice π_T . Now, suppose that the incoming covariances for a layer (ℓ) after t training steps has the form

$$(4.1) \quad \hat{C} = D + UKU^\top,$$

where D is a diagonal matrix, U has rank r , and K is a symmetric matrix. Note that the initial covariance matrix we start will already be in this form as it is the identity matrix. We think of D as the *bulk* part of the covariance and $UKU^\top$ as the *thin* part. The bulk D can carry most of the covariance, but, the thin part $UKU^\top$ contains the low dimensional correlations selected by the training.

We will now explain how we can compress!⁹ the information contained in the full covariance matrix using advice π to make the covariance matrix look like (4.1). Once again, for a fixed T , all the $O(T)$ terms can be suppressed and hence we don’t see any improvement (except potentially on wall-clock time) compressing the covariance information.

We are going to use an inductive argument. We already saw the the initial covariance matrix is already of the form (4.1). So, suppose that an incoming covariance from in a layer (ℓ) has the form (4.1). By suppressing (ℓ) and the training time t in the notation, assume that the weight matrix at that layer has the form

$$W = S + H, \quad S = PR^\top, \quad H = P_{A^\perp} G P_{B^\perp},$$

Then

$$W \hat{C} W^\top = W U K U^\top W^\top + W D W^\top.$$

The first term is already low-dimensional:

$$W U K U^\top W^\top = (W U) K (W U)^\top.$$

We therefore propagate it exactly. For the bulk term, expand

$$W D W^\top = H D H^\top + H D S^\top + S D H^\top + S D S^\top.$$

Since $S = PR^\top$, define

$$M := R^\top D R, \quad X := H D R.$$

⁸There is a nice way to make “exposed to training” precise. This is same as saying that the functions/operators are $\mathcal{F}_{\leq t}$ measurable in the sense of measure theory.

⁹In [2], ARC already highlights compression as an approach to *Matching Sampling Principle*. Here I see the advice and compression of the covariance as two sides of the same coin as compressed covariance provides information to the advice and vice versa.

Then

$$HDS^\top = XP^\top, \quad SDH^\top = PX^\top, \quad SDS^\top = PMP^\top.$$

Every term involving the explained core S is therefore again low-dimensional and can be retained exactly. Moreover, Y need not be computed by forming the residual matrix H . Indeed,

$$X = HDR = W(DR) - P(R^\top DR) = W(DR) - PM.$$

Thus all structure-dependent terms can be computed using dense matrix-thin matrix products. The only genuinely high-dimensional term remaining is

$$HDH^\top.$$

This is precisely where we use the conditional Gaussian law of the unexplored reservoir H . Let the initialization variance be

$$G_{ij} \sim \mathcal{N}(0, 1/n).$$

Conditional on the advice π ,

$$\mathbb{E}_G[HDH^\top] = \lambda P_{A^\perp}, \quad \lambda = \frac{1}{n} \text{tr}(P_{B^\perp} D).$$

We retain this conditional mean, and we also retain the exact diagonal of $HDH^\top$. Equivalently, we replace $HDH^\top$ by

$$\lambda P_{A^\perp} + \text{Diag}(\text{diag } HDH^\top - \lambda P_{A^\perp}).$$

Hence the only discarded quantity is $\text{OffDiag}(HDH^\top - \lambda P_{A^\perp})$. Since $P_{A^\perp} = I - Q_A Q_A^\top$, the new covariance matrix consists of a diagonal piece and a rank- $O(T)$ correction. Thus the output covariance again has the form

$$D' + U' K' U'^\top,$$

which is again in the desired form (4.1) and hence this closes the induction.

We can now choose some convenient choices of the matrices U' and K' for the purpose of our algorithm. We can let

$$U' = [WU, X, P, Q_A],$$

with coefficient matrix

$$K' = \begin{pmatrix} K & 0 & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & I & M & 0 \\ 0 & 0 & 0 & -\lambda I \end{pmatrix}.$$

To obtain the exact diagonal of the compressed covariance, set

$$d'_i = (HDH^\top)_{ii} + \lambda(Q_A Q_A^\top)_{ii}.$$

Indeed, the diagonal contribution $-\lambda Q_A Q_A^\top$ in the thin state cancels the second term, leaving precisely $(HDH^\top)_{ii}$ on the diagonal. Thus a trained linear layer maps $(m, D, U, K) \mapsto (m', D', U', K')$ without ever constructing a dense covariance matrix. For a general incoming thin feature dimension r , the linear update preserves the bulk-thin form and adds only $O(T)$ new advice-selected directions. In the non-tensorized variant mentioned in Remark 1.2 in the fixed- T regime one has $r = O(T)$, giving

$$O_L(n^2 T + n T^2 + T^3)$$

arithmetic. For the tensorized algorithm in the Section 5.1, the growth of r is controlled separately.

5. BEYOND FIRST-HERMITE PROPAGATION: TENSORIZED NONLINEAR UPDATES

For the nonlinear layers we essentially follow the algorithm of Wu et al. [3]. However, because of the form of the covariance matrix we are retaining from (4.1), we have the liberty to store higher order terms in the Hermite expansions¹⁰.

We start with a rough introduction to the algorithm of Wu et al [3] here. As before, we abuse notation to get rid of (ℓ) sub/superscripts for the matrices for a layer ℓ . In the Section 4, we saw that in the linear step we have a mean vector m and a bulk-thin covariance

$$C = D + UKU^\top.$$

Let $v_i = C_{ii}$ and let $Z_i \sim \mathcal{N}(m_i, v_i)$. We compute the Gaussian marginal moments

$$a_i = \mathbb{E}[\phi(Z_i)], \quad b_i = \mathbb{E}[\phi(Z_i)^2].$$

Thus the outgoing mean and marginal variance are

$$m_i^+ = a_i, \quad C_{ii}^+ = b_i - a_i^2.$$

For a matched bivariate Gaussian, the exact off-diagonal covariance has the Hermite expansion

$$\text{Cov}(\phi(Z_i), \phi(Z_j)) = \sum_{k=1}^{\deg \phi} \frac{C_{ij}^k}{k!} \gamma_{k,i} \gamma_{k,j},$$

where

$$\gamma_{k,i} = \mathbb{E}_{Z \sim \mathcal{N}(m_i, v_i)}[\phi^{(k)}(Z)].$$

For fixed training horizon, the diffuseness estimates obtained from SGD imply

$$C_{ij} = \tilde{O}(n^{-1/2}), \quad i \neq j.$$

So, if we follow the the algorithm in [3][Algorithm 2 in Appendix B], we see that $k \geq 2$ are lower order, and one may retain only

$$C_{ij}^+ \approx \gamma_{1,i} \gamma_{1,j} C_{ij},$$

and this gives us the new matrix

$$U^+ = \text{Diag}(\gamma_1)U.$$

But, we can do much better than this¹¹. This is what we describe in the next subsection.

5.1. Tensorization.¹² The tensorized variant removes this particular obstruction. Let $P = UKU^\top$. Now, we make the following important observation. For every $k \geq 1$,

$$P^{\circ k} = U^{[k]} K^{\otimes k} U^{[k]\top},$$

where the i th row of $U^{[k]}$ is $U_i^{\otimes k}$. Therefore, its entire polynomial nonlinear image of the explanation-selected covariance remains factorized. So, define

$$V_k = \frac{1}{\sqrt{k!}} \text{Diag}(\gamma_k) U^{[k]}, \quad K_k = K^{\otimes k}.$$

Then the entire off-diagonal matched-Gaussian covariance is represented by

$$\sum_{k=1}^{\deg \phi} V_k K_k V_k^\top.$$

The new thin feature matrix is therefore

$$U^+ = [V_1, \dots, V_{\deg \phi}],$$

¹⁰we refer the reader to [1][Section 3, Appendix B] for basics of Hermite expansion and the covariance propagation algorithm.

¹¹The difficulty is that this argument relies on the training-created correlations remaining perturbatively small. For a training horizon $T = T(n)$ growing with width, obtaining such a bound uniformly in T appears substantially harder.

¹²In [2], the ARC teams mentions progress on two layered trained networks with a tensorization idea, which seems philosophically different from the tensorization we do here.

with

$$K^+ = \text{blkdiag}(K_1, \dots, K_{\deg \phi}).$$

Finally, the diagonal bulk is chosen so that the exact Gaussian marginal variance is preserved:

$$d_i^+ = b_i - a_i^2 - (U^+ K^+ U^{+\top})_{ii}.$$

Hence for the “explained” covariance we propagate its entire polynomial nonlinear image:

$$U \longrightarrow \{U, U^{[2]}, \dots, U^{[\deg \phi]}\}.$$

If the incoming thin feature dimension is r , then after an activation of degree d the new feature dimension is at most

$$\sum_{k=1}^d r^k = O_d((1+r)^d).$$

Since each trained linear layer introduces only $O(T)$ additional thin directions, iteration through the fixed-depth network gives a thin feature dimension

$$O_L((1+T)^D), \quad D := \prod_{\ell=1}^L \max\{1, \deg \phi_\ell\}.$$

Consequently, the leading dense matrix–thin matrix arithmetic is

$$O_L(n^2(1+T)^D).$$

The point of tensorization is therefore not to improve the fixed- T power of n . In that regime the higher Hermite terms are already perturbatively small. Rather, tensorization removes the first-Hermite truncation on the explanation-selected correlations: larger training-created correlations can still be represented exactly at the matched-Gaussian level provided that they remain confined to a sufficiently low-dimensional algebra.

6. PSEUDOCODE

Algorithm 1 Tensorized bulk-thin mean-covariance propagation

Require: trained weights $\mathbf{W}^{(1)}, \dots, \mathbf{W}^{(L+1)} \in \mathbb{R}^{n \times n}$, activation functions $\phi_1, \dots, \phi_L$, and causal advice π_T giving, for each layer,

$$S^{(\ell)} = P^{(\ell)} R^{(\ell)\top}, \quad Q_A^{(\ell)}, Q_B^{(\ell)}$$

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1:  $\boldsymbol{\mu} \leftarrow \mathbf{0} \in \mathbb{R}^n$ 
2:  $\mathbf{d} \leftarrow \mathbf{1} \in \mathbb{R}^n$ 
3:  $\mathbf{U} \leftarrow$  empty matrix,  $\mathbf{K} \leftarrow$  empty matrix
4: for  $\ell = 1, \dots, L + 1$  do
5:    $\mathbf{W} \leftarrow \mathbf{W}^{(\ell)}$ ,  $\mathbf{P} \leftarrow P^{(\ell)}$ ,  $\mathbf{R} \leftarrow R^{(\ell)}$ 
6:    $\boldsymbol{\mu} \leftarrow \mathbf{W}\boldsymbol{\mu}$ 
7:    $\mathbf{Y} \leftarrow \mathbf{W}\mathbf{U}$ 
8:    $\mathbf{M} \leftarrow \mathbf{R}^\top \text{Diag}(\mathbf{d})\mathbf{R}$ 
9:    $\mathbf{X} \leftarrow \mathbf{W}(\text{Diag}(\mathbf{d})\mathbf{R}) - \mathbf{P}\mathbf{M}$ 
10:   $\lambda \leftarrow \frac{1}{n} \left( \sum_{i=1}^n d_i - \text{tr} \left( Q_B^{(\ell)\top} \text{Diag}(\mathbf{d}) Q_B^{(\ell)} \right) \right)$ 
11:   $H_{ij} \leftarrow W_{ij} - (\mathbf{P}\mathbf{R}^\top)_{ij}$  when needed
12:   $d'_i \leftarrow \sum_{j=1}^n d_j H_{ij}^2 + \lambda \sum_a \left( Q_A^{(\ell)} \right)_{ia}^2$  for  $i = 1, \dots, n$ 
13:   $\mathbf{U} \leftarrow [\mathbf{Y}, \mathbf{X}, \mathbf{P}, Q_A^{(\ell)}]$ 
14:   $\mathbf{K} \leftarrow \begin{pmatrix} \mathbf{K} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -\lambda \mathbf{I} \end{pmatrix}$ 
15:   $\mathbf{d} \leftarrow \mathbf{d}'$ 
16:  if  $\ell \leq L$  then
17:     $v_i \leftarrow d_i + (\mathbf{U}\mathbf{K}\mathbf{U}^\top)_{ii}$  for  $i = 1, \dots, n$ 
18:     $a_i \leftarrow \mathbb{E}_{Z \sim \mathcal{N}(\mu_i, v_i)}[\phi_\ell(Z)]$  for  $i = 1, \dots, n$ 
19:     $b_i \leftarrow \mathbb{E}_{Z \sim \mathcal{N}(\mu_i, v_i)}[\phi_\ell(Z)^2]$  for  $i = 1, \dots, n$ 
20:    for  $k = 1, \dots, \deg(\phi_\ell)$  do
21:       $\beta_{k,i} \leftarrow \frac{1}{\sqrt{k!}} \mathbb{E}_{Z \sim \mathcal{N}(\mu_i, v_i)}[\phi_\ell^{(k)}(Z)]$  for  $i = 1, \dots, n$ 
22:       $\mathbf{V}_k \leftarrow \text{Diag}(\boldsymbol{\beta}_k) \mathbf{U}^{[k]}$ 
23:       $\mathbf{K}_k \leftarrow \mathbf{K}^{\otimes k}$ 
24:    end for
25:     $\mathbf{U} \leftarrow [\mathbf{V}_1, \dots, \mathbf{V}_{\deg(\phi_\ell)}]$ 
26:     $\mathbf{K} \leftarrow \text{blkdiag}(\mathbf{K}_1, \dots, \mathbf{K}_{\deg(\phi_\ell)})$ 
27:     $d_i \leftarrow b_i - a_i^2 - (\mathbf{U}\mathbf{K}\mathbf{U}^\top)_{ii}$  for  $i = 1, \dots, n$ 
28:     $\mu_i \leftarrow a_i$  for  $i = 1, \dots, n$ 
29:  end if
30: end for
31: return  $\boldsymbol{\mu}$ 

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